WORKED SOLUTION

STEP II 1989, Question 14

Mechanics • Equilibrium

STEP II 1989, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

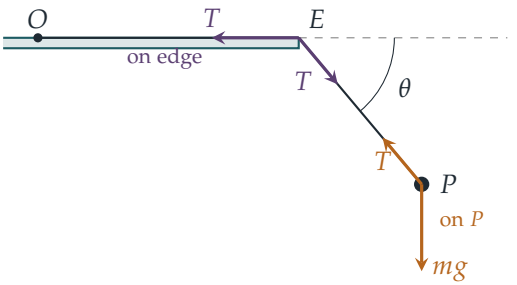
step.maths.org/questions/89-s2-q14

The key idea

Once the particle is released, the part of the string lying on the table stays put: it is held taut between the fixed end and the edge. So the edge acts as a fixed pivot, and the particle moves on a circle of radius a about it, exactly like a simple pendulum released from the horizontal. Energy gives the speed and the radial equation of motion gives the tension. The edge is smooth, so the tension is the same on both sides of it, and the force on the edge is the vector sum of two equal tensions pulling along the two pieces of string.

Solution

Model. The string is light and inextensible, the table-top is smooth and thin, and the particle is a point mass. Let O be the fixed end of the string and E the point of the edge where the string passes over it, so $OE = l - a$ and the hanging part EP has length a. Let θ be the angle the hanging part has turned through below the horizontal (the direction it started in, pointing away from the table).



The part OE stays on the table and stays still, since O is fixed and the string remains taut (we check below that $T > 0$). So P moves on a circle of centre E and radius a.

Speed. The table is smooth and the tension is always perpendicular to the motion of P, so neither does any work, and mechanical energy is conserved. When the string has turned through θ, P has fallen a height $a\sin\theta$, so starting from rest,

$$\tfrac{1}{2}mv^2 = mga\sin\theta \qquad\Longrightarrow\qquad v^2 = 2ga\sin\theta.$$

Tension. Resolving towards the centre E along PE: the tension acts along PE, and the component of the weight along PE is $mg\sin\theta$ *away* from E (for $0<\theta<\pi$, P is below the level of E). So

$$T - mg\sin\theta = \frac{mv^2}{a} = 2mg\sin\theta, \qquad \text{giving} \qquad \boxed{T = 3mg\sin\theta.}$$

This is positive for $0<\theta<\pi$, so the string stays taut throughout. At $\theta=\pi$ we have $v=0$: the particle just reaches the underside of the table, level with E, and swings back. So the motion takes place for $0\le\theta\le\pi$.

Force on the edge. The edge is smooth, so the tension is T on both sides of it. The string presses on the edge with two pulls of size T: one along EO, horizontal towards the fixed end, and one along EP. Take x horizontal away from the table and y vertically up. The unit vectors along EO and EP are $(-1,0)$ and $(\cos\theta,-\sin\theta)$, so the force on the edge is

$$\mathbf{F} = T\,(\cos\theta - 1,\ -\sin\theta).$$

Its magnitude is

$$|\mathbf{F}| = T\sqrt{(1-\cos\theta)^2+\sin^2\theta} = T\sqrt{2-2\cos\theta} = 2T\sin\tfrac{\theta}{2}$$

using $1-\cos\theta = 2\sin^2\frac{\theta}{2}$, and $\sin\frac{\theta}{2}\ge 0$ for $0\le\theta\le\pi$. (Geometrically, two equal forces T separated by angle $\pi-\theta$ add to $2T\cos\frac{\pi-\theta}{2}$, the same thing.) Substituting T and $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$,

$$|\mathbf{F}| = 6mg\sin\theta\sin\tfrac{\theta}{2} = 12mg\,s^2\sqrt{1-s^2}, \qquad \text{where } s=\sin\tfrac{\theta}{2}\in[0,1].$$

Ask yourself How do we maximise $s^2\sqrt{1-s^2}$ cleanly?

Since it is non-negative, it is enough to maximise its square, $h(u)=u^2(1-u)$ with $u=s^2\in[0,1]$. Then $h'(u)=2u-3u^2=u(2-3u)$, which is positive for $0<u<\frac{2}{3}$ and negative for $\frac{2}{3}<u\le 1$. So h has its maximum on $[0,1]$ at $u=\frac{2}{3}$, where $h=\frac{4}{9}\cdot\frac{1}{3}=\frac{4}{27}$. Hence the largest value of $s^2\sqrt{1-s^2}$ is $\sqrt{4/27}=\dfrac{2}{3\sqrt{3}}$, and

$$|\mathbf{F}|_{\max} = 12mg\cdot\frac{2}{3\sqrt{3}} = \frac{8mg}{\sqrt{3}},$$

as required. It occurs when $\sin^2\frac{\theta}{2}=\frac{2}{3}$, that is $\cos\theta = 1-2\sin^2\frac{\theta}{2} = -\frac{1}{3}$, so $\theta\approx 109.5^\circ$: the particle has already passed below the edge and is swinging back in under the table.

Checks

- At $\theta = \frac{\pi}{2}$ (string vertical) the speed is $\sqrt{2ga}$ and $T = 3mg$, the familiar result for a pendulum released from the horizontal. The force on the edge is then $T\sqrt{2} = 3\sqrt{2}\,mg \approx 4.24mg$, a little less than $8mg/\sqrt{3} \approx 4.62mg$, consistent with the maximum lying beyond the vertical.
- At $\theta = 0$ and $\theta = \pi$ the particle is momentarily at rest level with the edge, so $T = 0$ and the edge feels nothing, as the formula says.
- We integrated the equation of motion $\ddot{\theta} = (g/a)\cos\theta$ numerically: T from the radial equation matched $3mg\sin\theta$ to 10^{-12}, and the largest edge force came out at $\theta \approx 1.9106$ (that is, $\arccos(-\frac{1}{3})$) with value $4.6188\,mg$, matching $8/\sqrt{3}$.

Common slips

- Taking the force on the edge to be just T, or just the vertical part of the pull from the hanging string. The string on the table pulls the edge horizontally too, and both pulls must be added as vectors.
- Stopping the maximisation at $\theta \leq \frac{\pi}{2}$. The particle carries on under the table, and the maximum is in that later stage.

Beyond STEP

The tension here is a *constraint force*: it does no work and is whatever it needs to be to keep the particle on its circle. In Lagrangian mechanics such forces drop out of the equations of motion entirely, and are recovered afterwards (as we did with the radial equation) or by using Lagrange multipliers. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats circular motion, constraints and energy methods carefully at this level.

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